

estimate the median as the mean of the unique *two* central values. If the values x_j , $j = 0, \dots, N-1$, are sorted into ascending (or, for that matter, descending) order, then the formula for the median is

$$x_{\text{med}} = \begin{cases} x_{(N-1)/2}, & N \text{ odd} \\ \frac{1}{2}(x_{(N/2)-1} + x_{N/2}), & N \text{ even} \end{cases} \quad (14.1.14)$$

If a distribution has a strong central tendency, so that most of its area is under a single peak, then the median is an estimator of the central value. It is a more robust estimator than the mean is: The median fails as an estimator only if the area in the tails is large, while the mean fails if the first moment of the tails is large; it is easy to construct examples where the first moment of the tails is large even though their area is negligible.

To find the median of a set of values, one can proceed by sorting the set and then applying (14.1.14). This is a process of order $N \log N$. You might rightly think that this is wasteful, since it yields much more information than just the median (e.g., the upper and lower quartile points, the deciles, etc.). In fact, we saw in §8.5 that the element $x_{(N-1)/2}$ can be located in of order N operations. Consult that section for routines, including a method for getting a good approximation to the median in a single pass through the data.

The *mode* of a probability distribution function $p(x)$ is the value of x where it takes on a maximum value. The mode is useful primarily when there is a single, sharp maximum, in which case it estimates the central value. Occasionally, a distribution will be *bimodal*, with two relative maxima; then one may wish to know the two modes individually. Note that, in such cases, the mean and median are not very useful, since they will give only a “compromise” value between the two peaks.

CITED REFERENCES AND FURTHER READING:

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- Spiegel, M.R., Schiller, J., and Srinivasan, R.A. 2000, *Schaum's Outline of Theory and Problem of Probability and Statistics*, 2nd ed. (New York: McGraw-Hill).
- Stuart, A., and Ord, J.K. 1994, *Kendall's Advanced Theory of Statistics*, 6th ed. (London: Edward Arnold) [previous eds. published as Kendall, M., and Stuart, A., *The Advanced Theory of Statistics*], vol. 1, §10.15
- Norusis, M.J. 2006, *SPSS 14.0 Guide to Data Analysis* (Englewood Cliffs, NJ: Prentice-Hall).
- Chan, T.F., Golub, G.H., and LeVeque, R.J. 1983, “Algorithms for Computing the Sample Variance: Analysis and Recommendations,” *American Statistician*, vol. 37, pp. 242–247.[1]
- Cramér, H. 1946, *Mathematical Methods of Statistics* (Princeton, NJ: Princeton University Press), §15.10.[2]

14.2 Do Two Distributions Have the Same Means or Variances?

Not uncommonly we want to know whether two distributions have the same mean. For example, a first set of measured values may have been gathered before

some event, a second set after it. We want to know whether the event, a “treatment” or a “change in a control parameter,” made a difference.

Our first thought is to ask “how many standard deviations” one sample mean is from the other. That number may in fact be a useful thing to know. It does relate to the strength or “importance” of a difference of means *if that difference is genuine*. However, by itself, it says nothing about whether the difference *is* genuine, that is, statistically significant. A difference of means can be very small compared to the standard deviation, and yet very significant, if the number of data points is large. Conversely, a difference may be moderately large but not significant, if the data are sparse. We will be meeting these distinct concepts of *strength* and *significance* several times in the next few sections.

A quantity that measures the significance of a difference of means is not the number of standard deviations that they are apart, but the number of so-called *standard errors* that they are apart. The standard error of a set of values measures the accuracy with which the sample mean estimates the population (or “true”) mean. Typically the standard error is equal to the sample’s standard deviation divided by the square root of the number of points in the sample.

14.2.1 Student’s *t*-Test for Significantly Different Means

Applying the concept of standard error, the conventional statistic for measuring the significance of a difference of means is termed *Student’s t*. When the two distributions are thought to have the same variance, but possibly different means, then Student’s *t* is computed as follows: First, estimate the standard error of the difference of the means, s_D , from the “pooled variance” by the formula

$$s_D = \sqrt{\frac{\sum_{i \in A} (x_i - \bar{x}_A)^2 + \sum_{i \in B} (x_i - \bar{x}_B)^2}{N_A + N_B - 2}} \left(\frac{1}{N_A} + \frac{1}{N_B} \right) \quad (14.2.1)$$

where each sum is over the points in one sample, the first or second; each mean likewise refers to one sample or the other; and N_A and N_B are the numbers of points in the first and second samples, respectively. Second, compute t by

$$t = \frac{\bar{x}_A - \bar{x}_B}{s_D} \quad (14.2.2)$$

Third, evaluate the p -value or significance of this value of t for Student’s distribution with $N_A + N_B - 2$ degrees of freedom, by equation (6.14.11).

The p -value is a number between zero and one. It is the probability that $|t|$ could be this large or larger just by chance, for distributions with equal means. Therefore, a small numerical value of the p -value (0.01 or 0.001) means that the observed difference is “very significant.” The function $A(t|\nu)$ in equation (6.14.11) is one minus the p -value.

As a routine, we have

```
void ttest(VecDoub_I &data1, VecDoub_I &data2, Doub &t, Doub &prob)
```

stattests.h

Given the arrays data1[0..n1-1] and data2[0..n2-1], returns Student’s t as t , and its p -value as $prob$, small values of $prob$ indicating that the arrays have significantly different means. The data arrays are assumed to be drawn from populations with the same true variance.

```
{
    Beta beta;
```

```

    Doub var1,var2,svar,df,ave1,ave2;
    Int n1=data1.size(), n2=data2.size();
    avevar(data1,ave1,var1);
    avevar(data2,ave2,var2);
    df=n1+n2-2;
    svar=((n1-1)*var1+(n2-1)*var2)/df;
    t=(ave1-ave2)/sqrt(svar*(1.0/n1+1.0/n2));
    prob=beta.betai(0.5*df,0.5,df/(df+t*t));
}

```

Degrees of freedom.
Pooled variance.
See equation (6.14.11).

which makes use of the following routine for computing the mean and variance of a set of numbers,

```

moment.h void avevar(VecDoub_I &data, Doub &ave, Doub &var) {
    Given array data[0..n-1], returns its mean as ave and its variance as var.
    Doub s,ep;
    Int j,n=data.size();
    ave=0.0;
    for (j=0;j<n;j++) ave += data[j];
    ave /= n;
    var=ep=0.0;
    for (j=0;j<n;j++) {
        s=data[j]-ave;
        ep += s;
        var += s*s;
    }
    var=(var-ep*ep/n)/(n-1);
}

```

Corrected two-pass formula (14.1.8).

The next case to consider is where the two distributions have significantly different variances, but we nevertheless want to know if their means are the same or different. (A treatment for baldness has caused some patients to *lose* all their hair and turned others into werewolves, but we want to know if it helps cure baldness *on the average*!) Be suspicious of the unequal-variance *t*-test: If two distributions have very different variances, then they may also be substantially different in shape; in that case, the difference of the means may not be a particularly useful thing to know.

To find out whether the two data sets have variances that are significantly different, you use the *F*-test, described later on in this section.

The relevant statistic for the unequal-variance *t*-test is

$$t = \frac{\bar{x}_A - \bar{x}_B}{[\text{Var}(x_A)/N_A + \text{Var}(x_B)/N_B]^{1/2}} \quad (14.2.3)$$

This statistic is distributed *approximately* as Student's *t* with a number of degrees of freedom equal to

$$\frac{\left[\frac{\text{Var}(x_A)}{N_A} + \frac{\text{Var}(x_B)}{N_B} \right]^2}{\frac{[\text{Var}(x_A)/N_A]^2}{N_A - 1} + \frac{[\text{Var}(x_B)/N_B]^2}{N_B - 1}} \quad (14.2.4)$$

Expression (14.2.4) is in general not an integer, but equation (6.14.11) doesn't care.

The routine is

```
void ttest(VecDoub_I &data1, VecDoub_I &data2, Doub &t, Doub &prob) {
```

stattests.h

Given the arrays data1[0..n1-1] and data2[0..n2-1], this routine returns Student's t as t , and its p -value as $prob$, small values of $prob$ indicating that the arrays have significantly different means. The data arrays are allowed to be drawn from populations with unequal variances.

```
    Beta beta;
    Doub var1,var2,df,ave1,ave2;
    Int n1=data1.size(), n2=data2.size();
    avevar(data1,ave1,var1);
    avevar(data2,ave2,var2);
    t=(ave1-ave2)/sqrt(var1/n1+var2/n2);
    df=SQR(var1/n1+var2/n2)/(SQR(var1/n1)/(n1-1)+SQR(var2/n2)/(n2-1));
    prob=beta.betai(0.5*df,0.5,df/(df+SQR(t)));
}
```

Our final example of a Student's t -test is the case of *paired samples*. Here we imagine that much of the variance in *both* samples is due to effects that are point-by-point identical in the two samples. For example, we might have two job candidates who have each been rated by the same ten members of a hiring committee. We want to know if the means of the ten scores differ significantly. We first try `ttest` above, and obtain a value of $prob$ that is not especially significant (e.g., > 0.05). But perhaps the significance is being washed out by the tendency of some committee members always to give high scores and others always to give low scores, which increases the apparent variance and thus decreases the significance of any difference in the means. We thus try the paired-sample formulas,

$$\text{Cov}(x_A, x_B) \equiv \frac{1}{N-1} \sum_{i=0}^{N-1} (x_{Ai} - \bar{x}_A)(x_{Bi} - \bar{x}_B) \quad (14.2.5)$$

$$s_D = \left[\frac{\text{Var}(x_A) + \text{Var}(x_B) - 2\text{Cov}(x_A, x_B)}{N} \right]^{1/2} \quad (14.2.6)$$

$$t = \frac{\bar{x}_A - \bar{x}_B}{s_D} \quad (14.2.7)$$

where N is the number in each sample (number of pairs). Notice that it is important that a particular value of i label the corresponding points in each sample, that is, the ones that are paired. The p -value for the t statistic in (14.2.7) is evaluated for $N-1$ degrees of freedom.

The routine is

```
void tpctest(VecDoub_I &data1, VecDoub_I &data2, Doub &t, Doub &prob) {
```

stattests.h

Given the paired arrays data1[0..n-1] and data2[0..n-1], this routine returns Student's t for paired data as t , and its p -value as $prob$, small values of $prob$ indicating a significant difference of means.

```
    Beta beta;
    Int j, n=data1.size();
    Doub var1,var2,ave1,ave2,sd,df,cov=0.0;
    avevar(data1,ave1,var1);
    avevar(data2,ave2,var2);
    for (j=0;j<n;j++) cov += (data1[j]-ave1)*(data2[j]-ave2);
    cov /= (df=n-1);
    sd=sqrt((var1+var2-2.0*cov)/n);
    t=(ave1-ave2)/sd;
    prob=beta.betai(0.5*df,0.5,df/(df+t*t));
}
```